Lecture 6: Regression Analysis — Exercise Solutions

L2 - Statistics

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1 Your Turn #1: Getting to know the data

1.1 Question 1: Load the data

```r
library(haven)
grades <- read_dta("https://www.dropbox.com/s/wwp2cs9f0dubmhr/grade5.dta?dl=1")
```

1.2 Question 2: Describe the dataset

```r
# Unit of observation and number of observations
nrow(grades)
```

```
[1] 2019
```

```
ncol(grades)
```

```
[1] 9
```

```
names(grades)
```

```
[1] "schlcode"          "school_enrollment" "grade"
[4] "classize"          "avgmath"           "avgverb"
[7] "disadvantaged"     "female"            "religious"
```

Each row corresponds to **one class** in a Jewish public elementary school in Israel in 1991. There are **2019 classes** in the dataset.

Key variables: `classize` = number of students in the class; `avgmath` = average maths test score (0–100); `avgverb` = average verbal (reading) test score (0–100); `disadvantaged` = % of students from disadvantaged backgrounds; `religious` = 1 if the school is religious.

```
library(skimr)
grades %>%
  select(classize, avgmath, avgverb) %>%
  skim()
```

Table 1: Data summary

Name	Piped data
Number of rows	2019
Number of columns	3
Column type frequency:	
numeric	3
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
classize	0	1	29.93	6.56	5.00	26.00	31.00	35.00	44.00	
avgmath	0	1	67.30	9.60	27.69	61.12	67.82	74.08	93.93	
avgverb	0	1	74.38	7.68	34.80	69.83	75.38	79.84	93.86	

Key takeaways:

- Average class size: ~29 students, ranging from 5 to 45.
- Average math score: ~67; average verbal score: ~68, both out of 100.
- Both scores have substantial variation (sd 14).

1.3 Question 3: Prior intuitions and first visual insight

Prior intuition: Most people (and educational research) would expect *smaller* classes to be associated with *better* performance — more teacher attention, easier classroom management, etc.

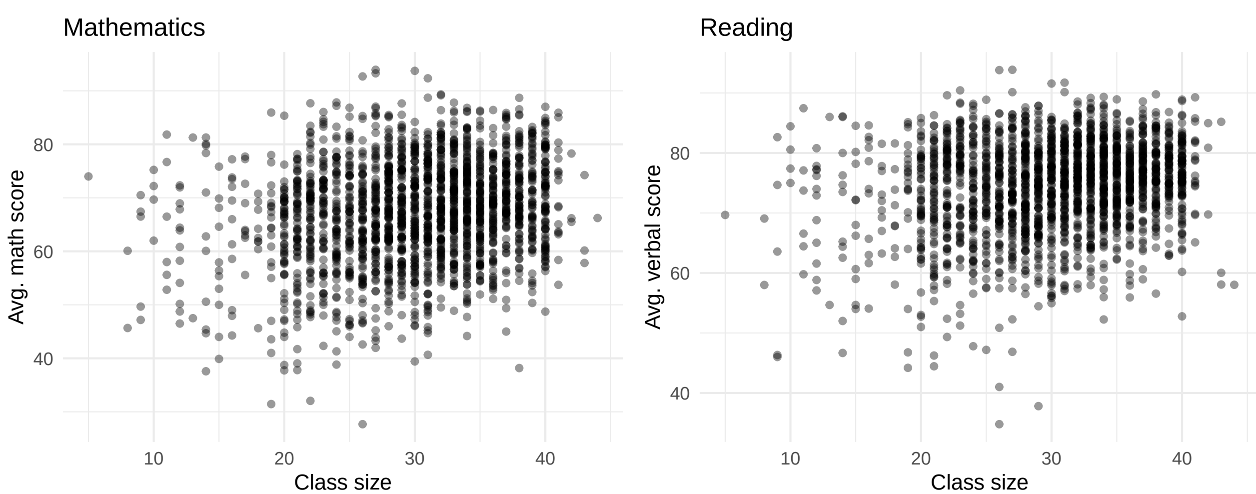
First insight: A scatter plot of class size against test scores is the natural starting point.

```
library(patchwork)

p_math <- ggplot(grades, aes(x = classize, y = avgmath)) +
  geom_point(alpha = 0.4, size = 1.5) +
  labs(x = "Class size", y = "Avg. math score", title = "Mathematics") +
  theme_minimal(base_size = 12)

p_verb <- ggplot(grades, aes(x = classize, y = avgverb)) +
  geom_point(alpha = 0.4, size = 1.5) +
  labs(x = "Class size", y = "Avg. verbal score", title = "Reading") +
  theme_minimal(base_size = 12)

p_math + p_verb
```



Both scatter plots show a weak, slightly positive cloud of points — **contrary to our prior intuition**.

1.4 Question 4: Correlations

```
cor(grades$classize, grades$avgmath, use = "complete.obs")
```

```
[1] 0.2171555
```

```r
cor(grades$classize, grades$avgverb, use = "complete.obs")
```

```
[1] 0.1899198
```

Both correlations are small and **positive** ($\approx +0.1$ to $+0.2$). The associations are weak, and in the "wrong" direction if we expected smaller classes to improve performance. This is almost certainly due to confounding: larger classes tend to be in wealthier urban schools that also have better-performing students for other reasons.

---

2 Your Turn #2: Simple OLS regression

2.1 Question 1: Regress `avgmath` on `classize`

```r
reg_math <- lm(avgmath ~ classize, grades)
reg_math
```

```

Call:
lm(formula = avgmath ~ classize, data = grades)

Coefficients:
(Intercept)     classize
    57.7939       0.3175
```

Intercept $\hat{\beta}_0 = 57.79$: The predicted average math score when class size is 0. Mathematically necessary but not meaningful (class size $= 0$ is impossible).

Slope $\hat{\beta}_1 = 0.317$: A 1-student increase in class size is associated, on average, with a 0.317-point **increase** in average math score. This positive sign reflects confounding, not a causal effect.

2.2 Question 2: Manual computation of coefficients

```r
b1_manual <- cov(grades$classize, grades$avgmath, use = "complete.obs") /
             var(grades$classize, na.rm = TRUE)
b0_manual <- mean(grades$avgmath, na.rm = TRUE) -
             b1_manual * mean(grades$classize, na.rm = TRUE)

cat("b1 (manual):", round(b1_manual, 4), "\n")
```

```
b1 (manual): 0.3175
```

```r
cat("b0 (manual):", round(b0_manual, 4), "\n")
```

```
b0 (manual): 57.7939
```

```r
cat("b1 from lm():", round(coef(reg_math)[2], 4), "\n")
```

```
b1 from lm(): 0.3175
```

```r
cat("b0 from lm():", round(coef(reg_math)[1], 4), "\n")
```

```
b0 from lm(): 57.7939
```

The manually computed coefficients match exactly.

2.3 Question 3: Predictions

```r
b0 <- coef(reg_math)[1]
b1 <- coef(reg_math)[2]

pred_25 <- b0 + b1 * 25
pred_35 <- b0 + b1 * 35

cat("Predicted avg. math score for class size 25:", round(pred_25, 2), "\n")
```

```
Predicted avg. math score for class size 25: 65.73
```

```r
cat("Predicted avg. math score for class size 35:", round(pred_35, 2), "\n")
```

```
Predicted avg. math score for class size 35: 68.91
```

2.4 Question 4: Verbal score regression

```r
reg_verb <- lm(avgverb ~ classize, grades)
reg_verb
```

```
Call:
lm(formula = avgverb ~ classize, data = grades)

Coefficients:
(Intercept)     classize
    67.7273       0.2223
```

```r
pred_verb_25 <- coef(reg_verb)[1] + coef(reg_verb)[2] * 25
pred_verb_35 <- coef(reg_verb)[1] + coef(reg_verb)[2] * 35
cat("Verbal score, class size 25:", round(pred_verb_25, 2), "\n")
```

```
Verbal score, class size 25: 73.28
```

```r
cat("Verbal score, class size 35:", round(pred_verb_35, 2), "\n")
```

```
Verbal score, class size 35: 75.51
```

The slope for verbal scores ($\hat{\beta}_1 \approx 0.222$) is similar in sign and magnitude to the math slope. Both suggest a weak positive association.

2.5 Question 5: Add regression line to scatter plot

```r
ggplot(grades, aes(x = classize, y = avgmath)) +
  geom_point(alpha = 0.35, size = 1.5) +
  geom_smooth(method = "lm", se = FALSE, color = "#E74C3C", linewidth = 1.5) +
  labs(x = "Class size", y = "Avg. math score",
       title = "OLS regression line: avg. math score on class size") +
  theme_minimal(base_size = 12)
```

3 Your Turn #3: Standard Errors

3.1 Questions 1–4: Computing $SE(\hat{\beta}_1)$ step by step

```
reg_reading <- lm(avgverb ~ classize, grades)

n     <- nrow(grades)
e     <- residuals(reg_reading)
SSR   <- sum(e^2)
SST_x <- sum((grades$classize - mean(grades$classize))^2)

sigma_hat <- sqrt(SSR / (n - 2))         # residual standard error
SE_b1     <- sigma_hat / sqrt(SST_x)     # SE of slope

cat("n           =", n, "\n")
```

```
n           = 2019
```

```
cat("SSR         =", round(SSR, 2), "\n")
```

```
SSR         = 114809.2
```

```r
cat("SST_x        =", round(SST_x, 2), "\n")
```

```
SST_x        = 86947.73
```

```r
cat("sigma_hat    =", round(sigma_hat, 4), "\n")
```

```
sigma_hat    = 7.5446
```

```r
cat("SE(b1) formula:", round(SE_b1, 5), "\n")
```

```
SE(b1) formula: 0.02559
```

```r
cat("SE(b1) lm():   ", round(summary(reg_reading)$coefficients[2, 2], 5), "\n")
```

```
SE(b1) lm():    0.02559
```

The formula and `lm()` give identical results.

3.2 Question 5: Effect of restricting x variation

Thinking question: Restricting to classes of size 20–25 drastically reduces the variation in `classize`. Since $SE(\hat{\beta}_1) = \hat{\sigma}/\sqrt{SST_x}$, a smaller SST_x directly inflates the SE: with almost no spread in x, the data contain very little information about the slope.

We can verify this empirically:

```r
grades_restricted <- grades %>% filter(classize >= 20 & classize <= 25)

reg_restricted <- lm(avgverb ~ classize, grades_restricted)

cat("SE(b1) full data:  ", round(summary(reg_reading)$coefficients[2, 2], 4), "\n")
```

```
SE(b1) full data:   0.0256
```

```r
cat("SE(b1) restricted: ", round(summary(reg_restricted)$coefficients[2, 2], 4), "\n")
```

```
SE(b1) restricted:  0.2575
```

The SE is **much larger** when x has less variation, confirming the formula.

---

4 Your Turn #4: Inference for verbal scores

4.1 Question 1: Run the regression and read `summary()`

```r
reg_verb <- lm(avgverb ~ classize, data = grades)
summary(reg_verb)
```

```

Call:
lm(formula = avgverb ~ classize, data = grades)

Residuals:
    Min      1Q  Median      3Q     Max 
-38.707  -4.414   0.756   5.350  20.303 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 67.72733    0.78395  86.392   <2e-16 ***
classize     0.22228    0.02559   8.688   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.545 on 2017 degrees of freedom
Multiple R-squared:  0.03607,   Adjusted R-squared:  0.03559 
F-statistic: 75.47 on 1 and 2017 DF,  p-value: < 2.2e-16
```

4.2 Question 2: Identify the key components

For the `classize` coefficient:

- **Estimate** $\hat{\beta}_1$: 0.2223 — a 1-student increase in class size is associated with a 0.2223-point increase in average verbal score.
- **SE**: 0.0256 — the estimated standard deviation of the sampling distribution of $\hat{\beta}_1$.
- t**-statistic**: 8.688 — the estimate is about 8.7 standard errors above zero.
- **p-value**: 7.5e-18 — very strong evidence against $H_0 : \beta_1 = 0$.

4.3 Question 3: Verify the t-statistic manually

```r
b1_est <- coef(reg_verb)[2]
se_est <- summary(reg_verb)$coefficients[2, 2]
df     <- nrow(grades) - 2

t_manual <- b1_est / se_est
```

```
cat("t-statistic (manual):", round(t_manual, 3), "\n")
```

```
t-statistic (manual): 8.688
```

```
cat("t-statistic (lm):    ", round(summary(reg_verb)$coefficients[2, 3], 3), "\n")
```

```
t-statistic (lm):     8.688
```

Both match.

4.4 Question 4: 95% Confidence Interval

```
t_crit_95 <- qt(0.975, df = df)   # 95% CI uses the 97.5th percentile (two-sided)

CI_low_95  <- b1_est - t_crit_95 * se_est
CI_high_95 <- b1_est + t_crit_95 * se_est

cat("95% CI: [", round(CI_low_95, 4), ",", round(CI_high_95, 4), "]\n")
```

```
95% CI: [ 0.1721 , 0.2725 ]
```

```
cat("Does the interval include 0?", CI_low_95 < 0 & CI_high_95 > 0, "\n")
```

```
Does the interval include 0? FALSE
```

The 95% CI does **not** include 0, consistent with rejecting $H_0 : \beta_1 = 0$ at the 5% significance level.

4.5 Question 5: Statistical significance and comparison with math

The p-value is far below 0.05, so the effect is statistically significant at the 5% level. The magnitude of the slope for verbal scores ($\hat{\beta}_1 \approx 0.222$) is similar to that for math scores ($\hat{\beta}_1 \approx 0.317$). Both are small in magnitude and positive — in the “wrong” direction relative to our prior expectation.

4.6 Question 6 (Bonus): Causal interpretation critique

The claim "larger classes *cause* better outcomes" is **incorrect** for at least two reasons:

1. **Regression gives association, not causation.** The positive coefficient simply tells us that larger classes tend to have higher average scores in this sample; it does not tell us *why*.
2. **Omitted variable bias.** A plausible confounder is school wealth or location: wealthier urban schools tend to have *both* larger classes *and* better-performing students (more resources, more educated parents). Class size therefore picks up part of the positive effect of school wealth on performance. Once we control for socioeconomic disadvantage, the coefficient on class size is expected to shrink or even turn negative.